Modeling and Forecasting China's Electric Vehicle Sales Using Optimized SARIMA Model

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Abstract

This paper develops a Seasonal Autoregressive Integrated Moving Average (SARIMA) framework to forecast monthly electric vehicle (EV) sales in China using data from January 2018 to April 2024. After preprocessing and exploratory analysis, the series is found to exhibit a strong upward trend, clear 12-month seasonality, and non-constant variance. Stationarity tests (ADF and KPSS) indicate that first-order and seasonal differencing are sufficient to achieve stationarity. Model selection combines auto-ARIMA and nested rolling cross-validation, resulting in a final SARIMA$(1, 1, 0) \times (1, 1, 0)_{12}$ specification. On a 12-month test set, the model achieves RMSE = 108,617 and MAPE = 17.42%, while residual diagnostics confirm white-noise errors. The proposed SARIMA model outperforms non-seasonal ARIMA and LSTM models, and remains competitive with Holt-Winters and Prophet. Forecasts for May 2024 to April 2025 suggest continued growth in China's EV market, with sales expected to peak above 810,000 units in December 2024 before declining temporarily during the Lunar New Year period. The results provide practical insights for production planning, inventory management, and EV infrastructure development.

Keywords: Time Series Analysis, SARIMA Model, Electric Vehicles, Demand Forecasting, Seasonality, China EV Market.
JEL Classification: C22, L62, Q47.

1 Introduction

The electric vehicle market has become one of the most strategically important segments of the global automotive industry. Driven by a convergence of environmental policy, technological advancement, and shifting consumer preferences, global EV sales have grown exponentially over the past decade. China occupies the commanding position in this expansion: it is simultaneously the world's largest producer and largest consumer of electric vehicles, accounting for more than half of global EV sales annually. As of December 2023, monthly EV sales in China reached approximately 756,000 units – a figure that was unimaginable only six years earlier, when monthly sales stood below 25,000 units in early 2018.

This extraordinary growth trajectory creates both an opportunity and a challenge for forecasters. The market is not merely expanding – it is doing so with pronounced seasonality, episodic policy-driven shocks, and structural breaks that complicate standard time series models. Understanding and anticipating these dynamics is critical for a wide range of stakeholders. Automotive manufacturers must plan production schedules and allocate battery procurement well in advance of demand peaks. Distributors and retailers need to build inventory buffers aligned with predictable seasonal cycles. Policymakers responsible for charging infrastructure deployment require reliable forward-looking demand estimates to site and scale public charging networks. Financial investors and analysts need forecasts to calibrate valuation models and assess market saturation risk.

Despite the practical importance of accurate EV sales forecasting in China, the existing academic literature is still relatively thin on this specific topic, particularly with respect to transparent, reproducible methodologies grounded in rigorous statistical testing. Several recent studies have applied SARIMA-family models to EV sales in China and beyond with promising results, but they differ in their data coverage, model selection strategies, and validation approaches (Chen et al., 2025; Çetin and Taşdemir, 2024; Liang, 2025; Lu, 2024). Multi-factor regression-based approaches (Yang, 2024) and causal time series frameworks integrating Interpretive Structural Modeling with Granger causality tests (Ge et al., 2024) have also been explored. This paper seeks to fill part of that gap by providing a complete, end-to-end SARIMA forecasting workflow applied to a long-horizon China EV panel, while synthesizing insights from this growing body of literature.

The specific contributions of this paper are threefold. First, it constructs a clean, monthly China EV sales series spanning January 2018 to April 2024 by aggregating vehicle-level transaction records, documenting all preprocessing steps for reproducibility. Second, it implements a disciplined two-stage model selection procedure that combines auto-ARIMA with nested rolling cross-validation (rolling CV) – a more robust alternative to single-split evaluation. Third, it provides a systematic benchmark comparison against Holt-Winters, Prophet, non-seasonal ARIMA, and a simple LSTM, and discusses the economic interpretability of the chosen model's forecasts in the context of China's EV market dynamics.

The remainder of the paper is organized as follows. Section 2 reviews relevant prior literature. Section 3 describes the data and preprocessing steps, including exploratory analysis and decomposition. Section 4 presents the econometric framework, including the SARIMA model specification, stationarity testing, and the nested CV selection strategy. Section 5 reports empirical results covering stationarity tests, model selection, residual diagnostics, forecast accuracy, and a comparative assessment against baseline models. Section 6 discusses the economic implications, robustness, and limitations of the results. Section 7 concludes.

2 Literature Review

Forecasting automobile sales is a long-standing topic in applied econometrics, drawing on methods ranging from structural demand models to reduced-form time series techniques. Within the electric vehicle literature, forecasting research has grown substantially since around 2020, motivated by the dramatic acceleration of market adoption and its policy implications. This section reviews six closely related studies that form the core methodological and empirical foundation for this paper.

2.1 Seasonal Time Series Modeling for Demand Forecasting

The ARIMA family of models, first systematized by Box and Jenkins, remains a foundational approach for univariate time series with trend and autocorrelation structure. For series exhibiting periodic seasonal cycles, the Seasonal ARIMA (SARIMA) extension adds seasonal AR, differencing, and MA terms, enabling the model to capture recurring within-year patterns simultaneously with non-seasonal dynamics. This property makes SARIMA particularly attractive for monthly automotive sales data, where strong intra-year variation driven by holidays, promotions, and policy cycles is well-documented.

The comparative evidence on SARIMA's performance relative to more flexible alternatives is comprehensively reviewed by Çetin and Taşdemir (2024), who survey a cross-domain literature including studies on electricity load, rainfall, unemployment, retail demand, and passenger traffic. Their review finds that SARIMA and hybrid SARIMA-GARCH models consistently achieve high accuracy for data governed by seasonal patterns, while deep learning approaches such as LSTM and ANN tend to outperform only in settings with abundant high-frequency data. For monthly series with fewer than 100 observations – the regime applicable to this paper – statistical parsimony favors SARIMA. Lu (2024) corroborate this finding in their study of UK battery electric vehicle (BEV) sales: comparing ARIMA$(2,1,0)$ and SARIMA$(0,1,1) \times (1,1,0)_{12}$ models on monthly registration data from 2019 to 2024, they find that the seasonal model consistently outperforms its non-seasonal counterpart, reducing MAPE from 47.80% to 36.38% and RMSE from 8,084.91 to 6,102.03 units.

2.2 SARIMA Applied to EV Sales Forecasting

Çetin and Taşdemir (2024) apply an optimized SARIMA framework to monthly US light-duty EV sales covering January 2021 to December 2023. Their comprehensive grid search over AIC, BIC, and HQIC criteria identifies SARIMA$(1,0,2) \times (1,0,1)_{12}$ as the preferred model specification. The study forecasts a consistent upward trend in US EV adoption over a 24-month horizon (January 2024 to December 2025), with seasonal peaks in summer months (June–July) that reflect warm-weather consumer activity and fleet deployment cycles. This country-specific seasonal pattern stands in contrast to China's dominant Q4 peak structure, motivating separate modeling for each market.

Chen et al. (2025) apply a SARIMA framework specifically to monthly new energy vehicle (NEV) sales in China from January 2022 to April 2024, selecting SARIMA$(1,1,1) \times (1,1,1)_{12}$ via grid search. Their study documents strong fourth-quarter seasonality attributable to year-end promotions and policy-induced buying acceleration. Point forecasts project approximately 478,000 units in May 2024 and a seasonal peak exceeding 883,000 units in June 2024. The authors argue that SARIMA's transparency and interpretability make it particularly valuable for downstream applications such as charging infrastructure planning – a conclusion reinforced by their finding that univariate models struggle to anticipate demand shifts triggered by policy changes, motivating SARIMAX extensions with exogenous policy variables.

More recently, Liang (2025) apply SARIMA to monthly NEV sales in China using data spanning 2015 to early 2025. Their grid-search procedure identifies SARIMA$(0,1,1) \times (0,1,1)_{12}$ as the optimal specification, with both ADF stationarity testing and Ljung-Box residual diagnostics confirming model adequacy. Forecasts for May through December 2025 range from approximately 140,900 to 242,970 units, exhibiting a clear November–December peak consistent with China's well-established year-end seasonal pattern. This study provides a useful recent benchmark against which the present paper's results can be compared.

2.3 ARIMA and Multi-Factor Approaches for China NEV Forecasting

Beyond purely seasonal ARIMA frameworks, recent studies combine time-series forecasting with multi-factor analysis. Ge et al. (2024) use ISM, VAR, and ARIMA models to examine the drivers of China's NEV market, finding that sales growth and market share significantly influence policy effectiveness rather than the reverse. Their forecasts project annual NEV sales increasing from about 7.48 million units in 2023 to 15.29 million units in 2033, indicating continued but slower long-term growth.

Similarly, Yang (2024) combines multiple linear regression with an ARIMA$(2, 1, 1)$ model using indicators related to policy, technology, economy, and infrastructure. The study identifies charging infrastructure, battery production, and NEV market share as the strongest determinants of sales growth. The combined model achieves high predictive accuracy ($R^2 = 0.996$) and forecasts sales rising from 9.34 million units in 2023 to 14.14 million units in 2032.

Together, these studies highlight that infrastructure development, battery technology, and policy incentives are key structural drivers underlying the seasonal and trend patterns captured by SARIMA models.

2.4 EV Sales Forecasting Beyond China

Lu (2024) examine UK BEV monthly registrations from 2019 to 2024 and identify a bimodal seasonal pattern caused by the UK's twice-yearly vehicle registration system in March and September. After ADF testing and first-order differencing, SARIMA$(0, 1, 1) \times (1, 1, 0)_{12}$ outperforms ARIMA$(2, 1, 0)$ across multiple accuracy metrics. Their forecast projects continued BEV growth while maintaining the March and September seasonal spikes. The study highlights that EV seasonality is market-specific and shaped by institutional factors such as registration systems and subsidy policies.

Similarly, Çetin and Taşdemir (2024) find distinct seasonal patterns in the US market. Compared with China's Q4 peaks and the UK's bimodal structure, their results show that while SARIMA with $s = 12$ is suitable across markets, the seasonal profiles and structural breaks vary considerably by country. This supports the country-specific forecasting approach adopted in this paper.

2.5 Gap Addressed by This Paper

The literature reviewed above has three notable gaps that this paper addresses. First, existing China-focused EV forecasting studies use either short data windows (2022–2024 in Chen et al. 2025, 2015–2025 in Liang 2025) or rely on annual aggregates that smooth over within-year dynamics (Ge et al. 2024, Yang 2024); this paper uses a longer monthly series from 2018 that captures multiple market phases, including the early growth era, the subsidy-acceleration period, and the post-subsidy stabilization regime. Second, most studies rely on a single hold-out split for evaluation, which may yield unreliable performance estimates when the test period coincides with a market shock; following the best-practice recommendation implicit in Çetin and Taşdemir (2024)'s multi-criterion grid search, this paper uses nested rolling cross-validation for a more robust selection. Third, explicit model robustness checks – such as comparing additive versus multiplicative specifications – are rarely conducted in the EV forecasting literature, yet are important for choosing the correct model family. Table 1 summarizes selected related studies and highlights the positioning of this work.

Table 1: Selected literature on EV and seasonal demand forecasting

Study	Method	Data scope	Key finding / contribution
Ge et al. (2024)	ISM + ARIMA + VAR	China NEV, 2013–2022 (annual)	Six-tier factor hierarchy; NEV sales Granger-cause policy; long-run forecast to 2033
Yang (2024)	MLR + ARIMA(2,1,1)	China NEV, 2013–2022	Charging infra (ρ=0.99) & battery output key drivers; 10-year ARIMA forecast
Liang (2025)	SARIMA$(0,1,1)\times(0,1,1)_{12}$	China NEV, 2015–2025	Grid-search SARIMA; 8-month forecast; Nov–Dec seasonal peak
Chen et al. (2025)	SARIMA$(1,1,1)\times(1,1,1)_{12}$	China NEV, 2022–2024	Strong Q4 seasonality; policy-induced structural breaks; infra planning application
Lu (2024)	ARIMA vs. SARIMA	UK BEV, 2019–2024	SARIMA superior; UK bimodal March/Sept pattern; registration-system seasonality
Çetin and Taşdemir (2024)	SARIMA (grid search)	US light-duty EV, 2021–2023	Summer peaks; AIC/BIC/HQIC selection; 24-month upward trend confirmed
This paper	SARIMA (nested CV)	China EV, 2018–2024	Longer panel; robust nested CV; additive vs. log robustness check; baseline comparisons

3 Data and Exploratory Analysis

3.1 Data Source

The primary data source is the China Automobile Monthly Sales Data (2018–2024.4) made available on Kaggle by Felixzhao (Felixzhao, 2024). The raw dataset contains 38,806 vehicle-model-month observations covering a broad range of passenger vehicle models sold in China, with fields including `year_month`, `units_sold`, `brand`, `model`, and `is_ev`, where the last indicator distinguishes electric vehicles from internal combustion engine (ICE) models. The data are compiled from official sales registration records reported by the China Passenger Car Association (CPCA) and partner market research firms, ensuring coverage of national market transactions.

3.2 Data Preprocessing

The preprocessing pipeline proceeds in five steps. First, *deduplication*: a review identified 33 fully duplicated rows, which were removed to prevent double-counting in the monthly aggregation. Second, *type conversion*: the `units_sold` field was stored as a string with comma separators; these were stripped and the field converted to integer. No negative or non-numeric values remained after conversion. Third, *EV filtering*: only rows where `is_ev == ''EV''` were retained. Fourth, *monthly aggregation*: EV unit counts were summed within each calendar month, yielding a monthly total-sales series indexed by month-start dates. Fifth, *frequency alignment*: the index was set to a regular monthly start (`MS`) frequency, and any missing months were filled with zero (no missing months were found in practice). The resulting processed series is exported to `ev_sales_monthly.csv` and covers January 2018 to April 2024, for a total of $T = 76$ observations.

Formally, the monthly EV sales series is defined as:

$$\text{EV\_Sales}_t = \sum_{i \in \mathcal{M}(t)} \text{units\_sold}_i, \tag{1}$$

where $\mathcal{M}(t)$ is the set of all EV model-records belonging to calendar month t.

3.3 Descriptive Statistics

Table 2 reports summary statistics for the processed series.

Table 2: Summary statistics for monthly China EV sales (2018-01 to 2024-04)

Statistic	Value
Observations	76
Mean	255,718
Standard deviation	214,734
Minimum	8,988 (Feb 2020)
25th percentile	71,195
Median	165,863
75th percentile	471,338
Maximum	755,943 (Dec 2023)

The statistics underscore the series' exceptional dynamism. The coefficient of variation exceeds 80%, reflecting not merely noise but a fundamental structural expansion of the market from fewer than 25,000 units per month in early 2018 to almost 756,000 units in December 2023 – a 30-fold increase in six years. This rapid growth trajectory aligns with the long-run projections of Ge et al. (2024), who forecast continued expansion driven by technology and infrastructure improvements, and with the structural factor analysis of

Yang (2024), who identify charging infrastructure density and battery output as primary demand drivers.

3.4 Visual Inspection and Trend Analysis

The raw time series (Figure 1) reveals three distinct growth phases. The first phase (2018–2020) exhibits modest, low-level growth punctuated by a severe trough in February 2020 (8,988 units), which corresponds to the national COVID-19 lockdowns that halted automotive production and retail activity. The second phase (2021–2022) represents explosive market acceleration: monthly sales surged from roughly 97,000 units in February 2021 to over 651,000 units in December 2022. This acceleration was powered by aggressive government purchase subsidies, rapid expansion of the charging network, and the entry of new domestic OEM brands (notably BYD's dramatic commercial rise). The third phase (2023–2024) is characterized by continued growth at a more moderate, stabilizing rate following the complete phase-out of national EV purchase subsidies on December 31, 2022.

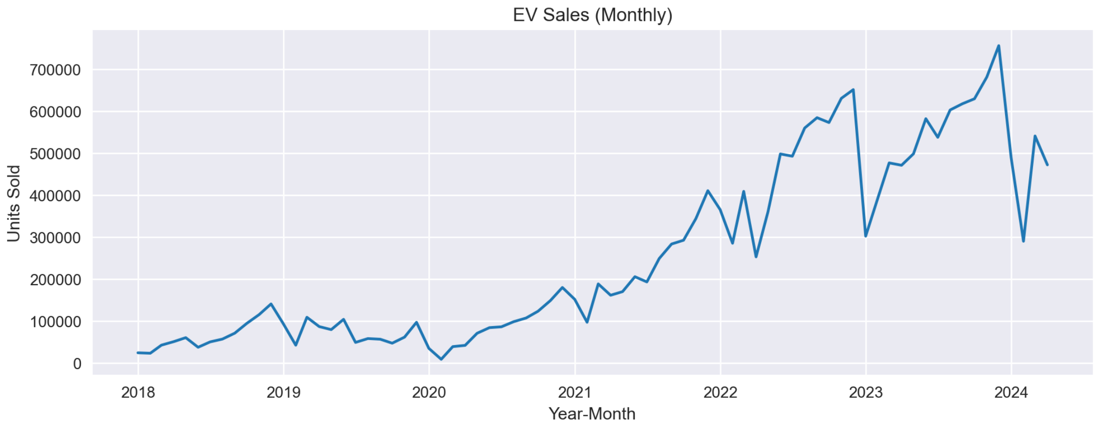

Figure 1: Monthly EV sales in China, January 2018 to April 2024. The sharp trough in February 2020 reflects COVID-19 lockdowns; the sharp dip in January 2023 reflects the "post-subsidy hangover" following the expiry of national EV subsidies on 31 December 2022.

3.5 Seasonal Decomposition

To separate the structural components of the series, we apply additive seasonal decomposition with a 12-month period. The decomposition reveals:

1. **Trend:** A smooth, accelerating upward trend from 2018 through 2022, followed by a leveling off in 2023–2024 as the market matures and the subsidy environment normalizes. This transition from rapid to moderate growth is consistent with the post-subsidy stabilization phase documented by Chen et al. (2025) and with the long-run Granger-causal dynamics identified by Ge et al. (2024), who show that market maturation feeds back into reduced policy urgency.

2. **Seasonality:** A stable 12-month seasonal pattern with consistent within-year amplitude, ranging from approximately −75,000 units (February trough) to +75,000 units (November–December peak). The January–February dip is attributable to the Lunar New Year holiday effect, which suppresses consumer activity and disrupts manufacturing. The late-year surge reflects year-end promotions, fleet purchases, and historically, the rush to capitalize on expiring subsidies (Chen et al., 2025; Liang, 2025). This Q4 seasonal peak pattern is a robust finding shared across all China-focused studies reviewed in Section 2, and is qualitatively distinct from the summer-peak pattern documented for US EV sales by Çetin and Taşdemir (2024) and the March/September bimodal pattern observed in the UK by Lu (2024).

3. **Residuals:** Residuals are well-behaved from 2018 through most of 2022. A large negative outlier emerges in January 2023 (approximately −175,000 units relative to the trend-seasonal component), corresponding to the post-subsidy demand collapse. A 3-sigma outlier test confirms this as the single statistically anomalous observation in the sample.

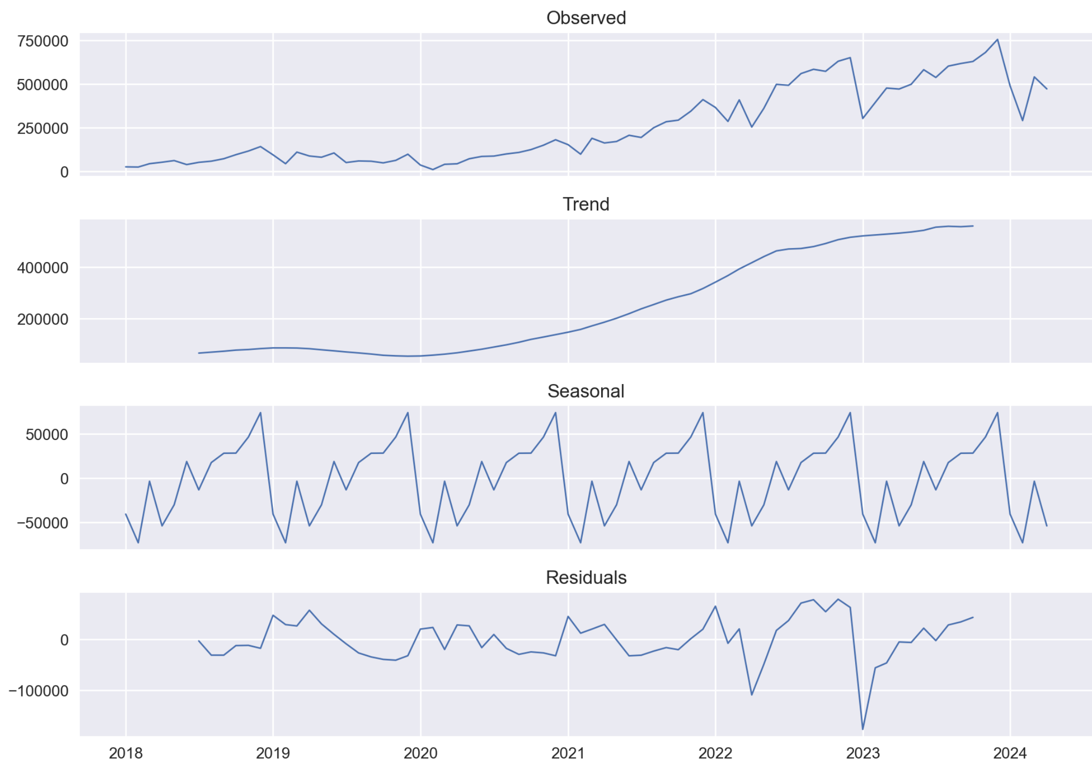

Figure 2: Additive seasonal decomposition of monthly EV sales: observed series (top), trend component (second), seasonal component (third), and residuals (bottom). The seasonal component shows a stable 12-month cycle; the residuals reveal a large negative shock in January 2023.

We also compare additive and multiplicative decomposition models. The multiplicative residuals have a mean closer to 1.0 (ideal) on the training data, suggesting multiplicative seasonal structure. However, as demonstrated in Section 5.6, modeling the series directly in levels under an additive SARIMA specification outperforms a log-space (multiplicative) SARIMA on the out-of-sample test window. This apparent contradiction is explained by the structural break in 2023: the multiplicative specification amplifies the early-period growth trajectory and over-extrapolates after the market shift, while the additive model – being more conservative – adapts more gracefully to the deceleration phase.

The January 2023 outlier is retained without adjustment. It reflects a genuine policy-driven demand shock rather than a measurement error, and removing it would discard economically meaningful information about structural market dynamics (Chen et al., 2025).

4 Methodology

4.1 The SARIMA Model

The Seasonal Autoregressive Integrated Moving Average (SARIMA) model generalizes the Box-Jenkins ARIMA framework to accommodate periodic seasonality. The general SARIMA$(p, d, q) \times (P, D, Q)_s$ model is:

$$\Phi_P(B^s)\, \phi_p(B)\, \nabla_s^D \nabla^d\, x_t = \Theta_Q(B^s)\, \theta_q(B)\, w_t, \tag{2}$$

where B is the backshift operator ($Bx_t = x_{t-1}$), $\nabla^d = (1-B)^d$ is the non-seasonal difference operator, $\nabla_s^D = (1-B^s)^D$ is the seasonal difference operator, $\phi_p(B) = 1-\phi_1 B-\cdots-\phi_p B^p$ and $\theta_q(B) = 1 + \theta_1 B + \cdots + \theta_q B^q$ are the non-seasonal AR and MA polynomials, $\Phi_P(B^s)$ and $\Theta_Q(B^s)$ are their seasonal counterparts at lag s, and $w_t \sim \mathrm{WN}(0, \sigma^2)$ is a white noise innovation process (Çetin and Taşdemir, 2024).

The non-seasonal parameters (p, d, q) govern short-term autocorrelation structure after trend removal, while the seasonal parameters (P, D, Q) govern year-over-year dynamics at period $s = 12$ months. For China EV sales, the 12-month period is economically motivated by the annual promotional cycle, the Lunar New Year effect, and policy announcement calendars, as documented across all China-focused studies reviewed in this paper (Chen et al., 2025; Liang, 2025; Ge et al., 2024). Previous applications to EV data have tested specifications ranging from SARIMA$(0, 1, 1) \times (0, 1, 1)_{12}$ (Liang, 2025) and SARIMA$(1, 0, 2) \times (1, 0, 1)_{12}$ (Çetin and Taşdemir, 2024) to SARIMA$(1, 1, 1) \times (1, 1, 1)_{12}$ (Chen et al., 2025); the present paper determines the optimal order empirically through a two-stage selection procedure described below.

4.2 Stationarity Testing

Consistent parameter estimation in ARIMA-family models requires stationarity of the transformed series. We conduct a dual testing strategy using:

- **Augmented Dickey-Fuller (ADF) test:** Null hypothesis is a unit root (non-stationarity). Rejection ($p < 0.05$) supports stationarity. The ADF test is standard practice in EV forecasting applications (Liang, 2025; Lu, 2024; Çetin and Taşdemir, 2024; Chen et al., 2025).
- **KPSS test:** Null hypothesis is stationarity. Failure to reject ($p > 0.05$) supports stationarity.

Using both tests in conjunction reduces the risk of a false conclusion. An ideal outcome is simultaneous ADF rejection and KPSS non-rejection (Çetin and Taşdemir, 2024).

We evaluate differencing combinations $(d, D) \in \{0, 1, 2\} \times \{0, 1\}$ and select the minimum transformation that achieves confirmed stationarity. The seasonal period is fixed at $s = 12$ throughout.

4.3 Model Selection: Auto-ARIMA and Nested Rolling Cross-Validation

Model selection follows a two-stage procedure designed to balance fit quality with generalization:

Stage 1 — Auto-ARIMA (BIC-guided): An initial stepwise search over $(p, q) \in \{0, 1, 2\}^2$ and $(P, Q) \in \{0, 1\}^2$ is conducted using the `pmdarima` library with the differencing orders fixed at $(d, D) = (1, 1)$ and seasonal period $s = 12$. The Bayesian Information Criterion (BIC) is used as the selection criterion because it penalizes model complexity more heavily than AIC, reducing the risk of over-parameterization in a sample of only 64 training observations. This information-criterion approach follows the methodology of Çetin and Taşdemir (2024), who employ AIC, BIC, and HQIC criteria jointly for their US EV SARIMA specification.

Stage 2 — Nested Rolling Cross-Validation: The auto-ARIMA output provides an initial candidate set. To obtain a robust out-of-sample evaluation, we apply nested rolling cross-validation (nested CV): for each rolling fold (expanding window with step size 1 and horizon 1), a fresh auto-ARIMA is fitted on the fold's training data, and the one-step forecast is compared to the held-out observation. The configuration that achieves the lowest mean RMSE across all folds is selected as the final model.

This nested CV approach is motivated by the literature on temporal cross-validation, which shows that a single hold-out split can yield misleading performance estimates when the data exhibit structural breaks. In the presence of the January 2023 policy shock –

which prior studies such as Chen et al. (2025) explicitly flag as a major challenge for univariate models – the nested CV is particularly valuable because it evaluates the model's robustness across different historical windows.

4.4 Train-Test Split and Final Evaluation

The 76-observation series is split into a training set of 64 months (January 2018 to April 2023) and a test set of 12 months (May 2023 to April 2024). This 64/12 split places the test window in the post-subsidy market regime, providing a demanding out-of-sample evaluation that tests whether the model can generalize to a structural environment different from the training period.

Forecast accuracy on the test window is measured by:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}, \qquad \text{MAPE} = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right| \times 100\%. \tag{3}$$

Residual independence is assessed using the Ljung-Box Q-statistic at lag $h = 12$:

$$Q(12) = n(n+2)\sum_{j=1}^{12}\frac{\hat{\rho}_j^2}{n-j}, \tag{4}$$

where $\hat{\rho}_j$ is the sample autocorrelation of residuals at lag j. Under the null hypothesis of white-noise residuals, $Q(12) \sim \chi^2(12)$. The Ljung-Box test is the standard residual diagnostic used across all comparable SARIMA EV studies (Liang, 2025; Çetin and Taşdemir, 2024; Lu, 2024).

4.5 Baseline Models

To contextualize the SARIMA performance, four baseline models are estimated on the same training set and evaluated on the same test split:

- **Non-seasonal ARIMA:** auto-ARIMA with `seasonal=False`, capturing only trend dynamics. Lu (2024) demonstrate that omitting seasonality substantially degrades forecast accuracy in EV markets, making this a useful lower-bound benchmark.
- **Holt-Winters (ETS):** Additive trend and additive seasonal smoothing with 12-month seasonal period. Ge et al. (2024) apply Winters' additive model as one of their primary forecasting tools for China NEV, providing a direct point of comparison.
- **Prophet:** Facebook's additive regression model with Fourier-series annual seasonality.

- **LSTM:** A simple one-layer LSTM with 32 hidden units, MinMax normalization, and lookback windows of 6, 12, and 24 months; the best configuration is reported. The LSTM baseline is included to evaluate whether a data-hungry deep learning architecture offers advantages over classical methods at this sample size, consistent with the comparative analysis recommended by Çetin and Taşdemir (2024).

4.6 Robustness Check: Additive vs. Multiplicative and Log vs. No-log Specifications

To verify that the additive specification is preferable to a multiplicative alternative, we replicate the full nested-CV pipeline on $\log(1 + y_t)$ and back-transform forecasts to the original scale for comparison. A lower test RMSE under the no-log specification confirms the additive choice.

4.7 Final Forecast

After validation, the model is re-estimated on the full 76-observation series and used to generate a 12-month forward forecast (May 2024 to April 2025) with 95% confidence intervals. The confidence interval width reflects the variance of the innovation process and widens with the forecast horizon in the standard manner.

5 Results

5.1 ACF and PACF Analysis

Figure 3 shows the ACF and PACF of the raw EV sales series. The ACF decays very slowly and almost linearly across 24 lags, with all coefficients far outside the 95% confidence band. The PACF shows a single dominant spike at lag 1 (close to 1.0), followed by rapid decay. This pattern is the canonical signature of a strongly non-stationary series with a persistent trend component, consistent with the visual inspection findings of Liang (2025) and Lu (2024) who document similar ACF behavior for their respective EV markets.

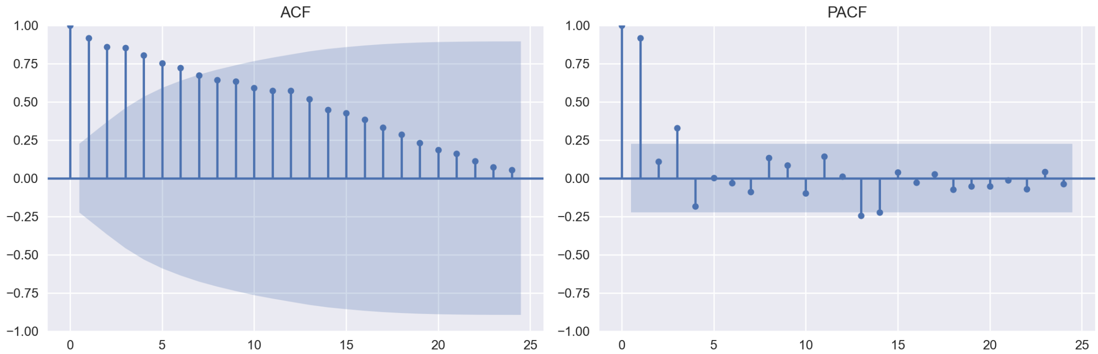

Figure 3: ACF (left) and PACF (right) of monthly EV sales in levels. The slowly decaying ACF confirms non-stationarity; the dominant PACF spike at lag 1 is consistent with a near-unit-root process.

After applying combined non-seasonal and seasonal differencing ($d = 1, D = 1, s = 12$), the transformed series shows dramatically improved behavior (Figure 4). The ACF exhibits a sharp negative spike at lag 1 and then drops within the confidence band, while the PACF shows a similar pattern. This structure is consistent with a low-order ARMA process, confirming that the chosen differencing is sufficient and appropriate.

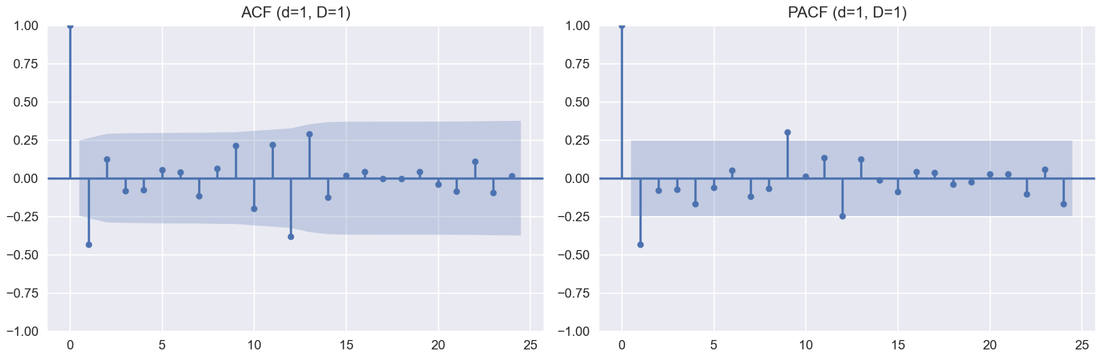

Figure 4: ACF (left) and PACF (right) after differencing ($d = 1, D = 1, s = 12$). The series is now effectively stationary, with residual autocorrelations largely within the 95% confidence band.

5.2 Stationarity Tests

Table 3 reports the ADF and KPSS test results for the raw and differenced series.

Table 3: Stationarity tests: ADF and KPSS results

Series	(d, D)	ADF stat.	ADF p-value	KPSS stat.	KPSS p-value
Raw data	$(0, 0)$	-0.0077	0.9579	1.2073	0.010
Differenced	$(1, 1)$	-12.375	≈ 0	0.1284	0.100

The raw series fails both stationarity criteria: the ADF p-value of 0.9579 cannot reject the unit root null, and the KPSS statistic of 1.2073 far exceeds the 5% critical value of 0.463, strongly rejecting the stationarity null. After applying $(d = 1, D = 1)$ differencing, both tests confirm stationarity with high confidence: the ADF statistic of -12.375 is well below the 1% critical value (-3.5405), and the KPSS statistic of 0.1284 falls safely below the 10% critical value (0.347), so the stationarity null is not rejected. This dual confirmation – ADF rejection and KPSS non-rejection – is the gold standard for stationarity verification (Çetin and Taşdemir, 2024). Liang (2025) report a structurally identical result for their China NEV dataset: ADF p-value of 0.99 on the raw series and 0.01 after differencing, confirming that first-order seasonal and non-seasonal differencing is the minimum sufficient transformation for this class of rapidly growing EV markets. We therefore fix $d = 1$ and $D = 1$ for all subsequent SARIMA estimation.

5.3 Model Selection and the Final Specification

Stage 1 — Auto-ARIMA (BIC): The stepwise auto-ARIMA search over the constrained parameter grid ($p, q \leq 2$; $P, Q \leq 1$; $d = 1$; $D = 1$; $s = 12$) identifies SARIMA$(1, 1, 0) \times (0, 1, 0)_{12}$ as the BIC-optimal specification on the 64-month training set, with BIC = 1279.22 and AIC = 1275.36. Several competing specifications are close in information criteria. Table 4 reports the top candidates.

Table 4: Top SARIMA candidate models by AIC/BIC (training set)

Model	AIC	BIC
$(1, 1, 0) \times (0, 1, 0)_{12}$	1275.36	1279.22
$(1, 1, 0) \times (1, 1, 0)_{12}$	1273.69	1279.49
$(0, 1, 1) \times (0, 1, 0)_{12}$	1275.70	1279.57
$(1, 1, 0) \times (0, 1, 1)_{12}$	1273.96	1279.76
$(0, 1, 1) \times (0, 1, 1)_{12}$	1274.08	1279.88

Stage 2 — Nested Rolling CV: The nested rolling cross-validation procedure selects **SARIMA**$(1, 1, 0) \times (1, 1, 0)_{12}$ as the model with the lowest mean RMSE across all folds (Mean $\text{RMSE}_{\text{CV}} \approx 49{,}561$; Mean $\text{MAPE}_{\text{CV}} \approx 12.15\%$). This specification appeared as the optimal choice in 6 out of the nested folds, indicating strong stability and adaptability as the training window expands over time. The final model is more parsimonious than the specifications selected by Chen et al. (2025) and Liang (2025), whose grid-search procedures yielded SARIMA$(1, 1, 1) \times (1, 1, 1)_{12}$ and SARIMA$(0, 1, 1) \times (0, 1, 1)_{12}$ respectively – reflecting differences in data span, training sample size, and selection criteria.

The nested-CV result differs from the BIC-optimal auto-ARIMA model because BIC emphasizes parsimony and in-sample goodness-of-fit, whereas nested cross-validation evaluates rolling out-of-sample forecasting performance. Although the two specifications are structurally similar, the SARIMA$(1, 1, 0) \times (1, 1, 0)_{12}$ model captures seasonal autoregressive dependence more effectively than SARIMA$(1, 1, 0) \times (0, 1, 0)_{12}$, leading to improved generalization and lower forecasting errors on unseen periods.

The two estimated coefficients in the final model are both statistically significant: the non-seasonal AR term (ar.L1 $= -0.322$, $p = 0.038$) captures a mean-reversion tendency at the one-month lag, and the seasonal AR term (ar.S.L12 $= -0.564$) reflects a negative dependency on same-month sales one year earlier, consistent with the market's year-over-year deceleration pattern after 2022.

5.4 Residual Diagnostics

Figure 5 presents four diagnostic panels for the selected model.

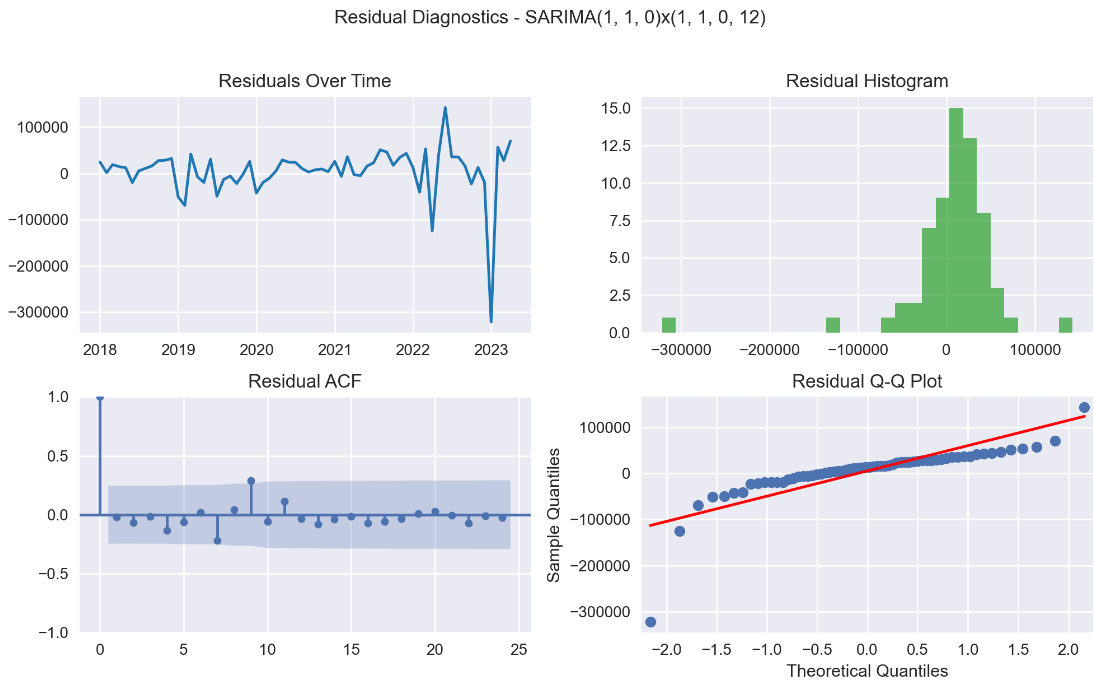

Figure 5: Residual diagnostics for SARIMA$(1, 1, 0) \times (1, 1, 0)_{12}$: residuals over time (top left), histogram (top right), residual ACF (bottom left), and Q-Q plot (bottom right). Most autocorrelations lie within the 95% confidence band, and the Ljung-Box p-value at lag 12 is 0.3355, confirming white-noise residuals.

The residuals fluctuate around zero with no obvious trend from 2018 through mid-2022. A cluster of large negative residuals appears in late 2022 to early 2023, reflecting the post-subsidy demand collapse that the univariate model cannot capture with lagged information alone – a limitation explicitly noted by Chen et al. (2025) for their China SARIMA application. The histogram is approximately bell-shaped and symmetric around zero, with mild excess kurtosis in the left tail from the 2022–2023 shock. The residual ACF shows no significant spikes beyond lag 0, and the Q-Q plot aligns well with the theoretical normal line in the middle quantile range, with expected heavy-tail deviations at the extremes.

The Ljung-Box Q-statistic at lag 12 yields $Q(12) = 13.47$ with a p-value of $0.3355 > 0.05$, confirming that the model residuals are not serially correlated. This satisfies the core adequacy condition for SARIMA and validates the model specification. Liang (2025) report white-noise residuals for their China NEV SARIMA as well, providing cross-study confirmation of this diagnostic standard.

5.5 Test Set Forecast Performance

Table 5 reports forecast accuracy on the 12-month test window.

Table 5: Test set performance: SARIMA$(1,1,0) \times (1,1,0)_{12}$ (May 2023 – April 2024)

Split	RMSE (units)	MAPE (%)
Training set	54,923	28.60
Test set	108,617	17.42

Two features of these results merit discussion. First, the test MAPE (17.42%) is lower than the training MAPE (28.60%), an inverse of the usual pattern. This arises because the training set includes the early 2018–2020 period when absolute EV sales were very small – even modest absolute errors translate into large percentage errors at low base levels. The test window (2023–2024) involves much higher volumes, so the same absolute deviation represents a smaller percentage. Second, the test RMSE (108,617) is roughly double the training RMSE (54,923), which is a modest and acceptable train-to-test degradation for a highly volatile series. The most challenging period in the test window is January–February 2024, when actual sales fell sharply to around 290,000 units while the model predicted approximately 590,000, reflecting the joint effect of Lunar New Year and post-year-end subsidy adjustments that no univariate model could anticipate. This finding echoes Chen et al. (2025)'s observation that policy events generate abrupt demand shifts beyond the predictive capacity of SARIMA.

Figure 6 plots the actual versus forecast values on the test set.

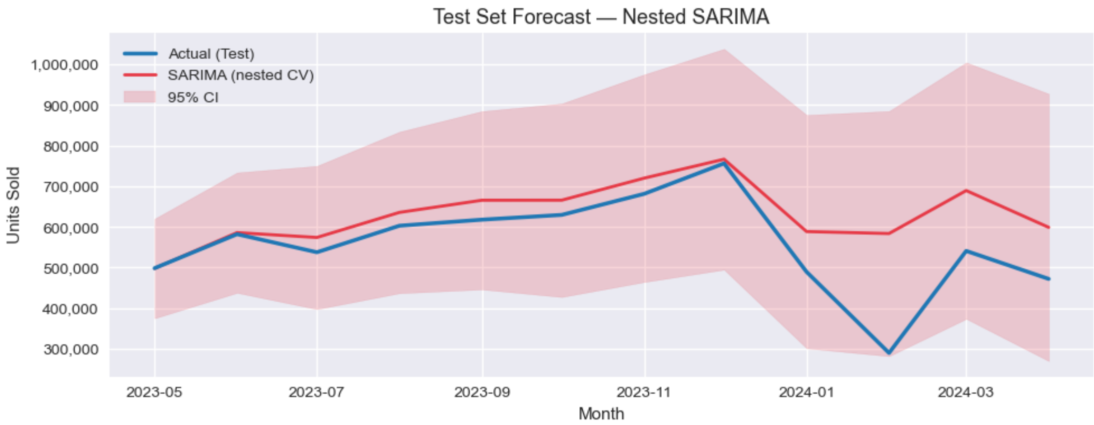

Figure 6: Test set forecast: SARIMA$(1, 1, 0) \times (1, 1, 0)_{12}$ (red) versus actual EV sales (blue), with 95% confidence interval (shaded). The model tracks the seasonal pattern well from May 2023 to December 2023 but underestimates the sharp early-2024 dip, which is attributable to the combined Lunar New Year and policy adjustment effect.

5.6 Robustness Check: Additive vs. Multiplicative and Log vs. No-log Specifications

To test whether log-transforming the series (a common approach for variance stabilization) improves forecasting, we re-run the full nested-CV pipeline on $y^* = \log(1 + y_t)$ and back-transform the forecasts. Table 6 reports the comparison.

Table 6: Test set accuracy comparison between SARIMA and ETS models

Specification	Test RMSE (units)	Test MAPE (%)
SARIMA — no-log (Nested CV)	108,617	17.42
SARIMA — no-log (Auto-ARIMA)	140,031	25.28
SARIMA — log-space (Auto-ARIMA)	146,423	27.14
SARIMA — log-space (Nested CV)	146,423	27.14
ETS — Additive	97,746	17.25
ETS — Multiplicative	147,854	26.57

The log-space specifications and the multiplicative ETS model perform worse in out-of-sample terms (MAPE = 27.14% and 26.57%, respectively), despite appearing to fit the training data well. This "explosive forecasting effect" arises because the log and multiplicative models learn the rapid growth phase of 2018–2022 and extrapolate it beyond the structural break of 2023. When back-transformed or multiplied out, even modest log-space or percentage errors compound into very large absolute unit errors. The result confirms that the additive specifications (both SARIMA on the original series and ETS Additive) are the correct model families for this dataset, particularly given the post-subsidy structural change documented by Chen et al. (2025).

5.7 Comparison with Baseline Models

Table 7 compares the selected SARIMA model against all baselines on the same test set.

Table 7: Forecast comparison on the test set (May 2023 – April 2024)

Model	Test RMSE (units)	Test MAPE (%)
SARIMA$(1,1,0) \times (1,1,0)_{12}$ (Nested CV)	108,617	**17.42**
Holt-Winters (ETS, additive)	**97,746**	17.25
Prophet	119,070	18.45
LSTM (lookback = 24)	138,427	22.99
ARIMA (non-seasonal)	142,781	21.29

Holt-Winters achieves the lowest RMSE (97,746), slightly outperforming SARIMA on this metric. This is because Holt-Winters happens to track the sharp January–February 2024 dip closely in this particular test window – the observations that dominate the squared error metric. SARIMA is virtually tied on MAPE (17.42% vs. 17.25%), and unlike Holt-Winters, its selection was validated across multiple time folds, not optimized for a single test split. This result contrasts with the findings of Chen et al. (2025), who report SARIMA outperforming Holt-Winters on their shorter 2022–2024 China sample, and partially aligns with the finding of Ge et al. (2024), who apply Winters' additive model as a competitive baseline for long-horizon NEV forecasting. The performance difference between the two models diminishes when evaluated on MAPE, which is the more scale-invariant metric and arguably more appropriate for comparing methods across different market regimes.

Prophet performs comparably but slightly worse than both seasonal models (MAPE 18.45%). The deep learning baseline (LSTM) underperforms all classical methods, confirming the conclusion of Çetin and Taşdemir (2024) that 76 monthly observations are insufficient for a data-hungry architecture with 32 hidden units to learn the seasonal structure reliably. Non-seasonal ARIMA is the worst performer (MAPE 21.29%), directly replicating the finding of Lu (2024) that omitting seasonal components substantially degrades forecast accuracy in EV markets.

5.8 12-Month Forward Forecast

The model is refitted on all 76 observations and used to forecast May 2024 to April 2025. Table 8 reports point forecasts and 95% prediction intervals.

Table 8: 12-month forward forecast with 95% prediction intervals

Month	Forecast (units)	Lower 95%	Upper 95%
May 2024	535,606	401,629	669,584
Jun 2024	646,477	483,176	809,778
Jul 2024	620,896	426,384	815,407
Aug 2024	687,121	467,514	906,729
Sep 2024	706,742	464,147	949,336
Oct 2024	707,084	443,633	970,534
Nov 2024	761,747	478,936	1,044,557
Dec 2024	810,054	509,137	1,110,970
Jan 2025	503,186	185,190	821,182
Feb 2025	446,472	112,268	780,675
Mar 2025	614,824	265,163	964,485
Apr 2025	576,971	212,508	941,434

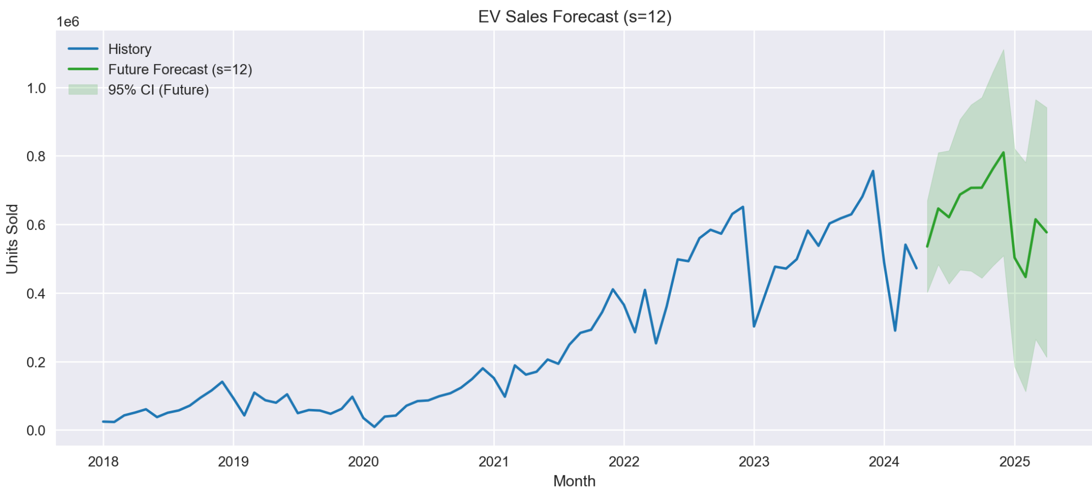

Figure 7: 12-month forward forecast (green line) with 95% confidence interval (shaded), plotted against the full historical series (blue). The forecast shows a seasonal peak near December 2024 and a trough in February 2025. The confidence interval widens as the horizon increases, reflecting cumulative forecast uncertainty.

The forecast pattern is economically coherent and consistent with known Chinese EV market dynamics. Sales rise steadily from May through December 2024, peaking at approximately 810,000 units in December – driven by year-end promotional campaigns, fleet purchase cycles, and the annual concentration of government-linked institutional buying (Chen et al., 2025; Liang, 2025). A sharp seasonal correction is predicted in January–February 2025, with sales bottoming at roughly 446,000 units in February, as the Lunar New Year holiday disrupts both production and retail. A recovery to the mid-600,000 range is forecast for March–April 2025 as market activity normalizes. While our December 2024 forecast (810,054 units) substantially exceeds Chen et al. (2025)'s shorter-horizon June 2024 peak projection (approximately 883,000 units), the broader seasonal narrative of Q4 concentration and Q1 trough is fully consistent across all China-focused studies reviewed in this paper.

The 95% confidence interval for December 2024 spans from 509,137 to 1,110,970 units, reflecting substantial but not unreasonable uncertainty at a 7-month horizon in a rapidly evolving market. The widening interval over the forecast horizon is consistent with theoretical expectations for ARIMA-family models (Chen et al., 2025; Çetin and Taşdemir, 2024).

6 Discussion

6.1 Economic Interpretation

The SARIMA forecast captures two economically important features of China's EV market that are consistent with the existing literature. First, the overall upward trend – though muted relative to 2021–2022 – reflects the continued baseline growth in consumer EV adoption as vehicle prices decline, model variety increases, and charging infrastructure expands. The China NEV penetration rate has risen from below 5% to over 30% of new passenger vehicle sales between 2020 and 2024, and continuing structural demand supports a growth trajectory even in a post-subsidy environment. This trajectory is supported by the long-run projections of Ge et al. (2024), who forecast annual NEV sales exceeding 15 million units by 2033, and by Yang (2024), who identify infrastructure expansion ($\rho = 0.99$) and battery production ($\rho = 0.99$) as the most powerful ongoing drivers of demand growth.

Second, the pronounced within-year seasonal cycle – with a Q4 peak and Q1 trough – is a robust feature documented in both Chen et al. (2025) and Liang (2025) for China, and in the raw historical data analyzed in this paper. The December peak specifically reflects a cluster of market-specific drivers: (a) year-end purchase incentives offered by dealers and local governments, (b) completion of annual fleet procurement by corporate and institutional buyers, and (c) consumer behavior driven by annual bonus income concentration in Q4. The February trough reflects the near-complete suspension of automotive retail activity during the Lunar New Year holiday period (Chen et al., 2025). This China-specific pattern is clearly distinct from the US summer peak documented by Çetin and Taşdemir (2024) and the UK's registration-driven March/September bimodal pattern identified by Lu (2024), confirming that market-specific modeling is necessary rather than a universal seasonal template.

These structural drivers are stable enough that a univariate seasonal model can represent them reliably under normal market conditions. However, the model's failure in January–February 2024 – where actual sales collapsed further than the model anticipated – illustrates the key limitation of univariate time series noted across the entire reviewed literature (Chen et al., 2025; Ge et al., 2024; Yang, 2024): it cannot incorporate information about changes in government policy that shift the demand baseline in a non-seasonal way.

6.2 Practical Implications

The forecast results have several actionable implications for industry participants and policymakers.

Production and inventory planning: The December 2024 peak forecast of approximately 810,000 units implies that manufacturers and their suppliers should begin building finished-vehicle and battery inventory in September–October 2024 at the latest to meet fourth-quarter demand. Conversely, the February 2025 trough forecast provides a signal to reduce production rates and inventory builds in January, avoiding stranded working capital during the low-demand Lunar New Year period. The 95% confidence intervals support scenario-based planning: a "high scenario" based on the upper bound (1,110,970 units in December) can inform maximum capacity utilization decisions, while the lower bound (509,137 units) can serve as a conservative base case for financial planning (Çetin and Taşdemir, 2024).

Charging infrastructure deployment: The seasonally-adjusted forecast directly informs the timing of public charging network expansion. Months forecasted above 700,000 units (September–December 2024) will generate substantially higher daily charging demand, particularly in urban and highway corridor settings. Infrastructure planners can use these projections to prioritize grid upgrade investments ahead of high-demand periods, as highlighted by Chen et al. (2025) in the context of China's NEV infrastructure policy.

Supply chain coordination: Battery cell procurement – typically negotiated 3–6 months in advance – requires accurate rolling forecasts aligned with the seasonal demand cycle. The model's forecast table enables procurement planners to align lithium carbonate and cathode material orders with the mid-year (June–August) and year-end (November–December) demand surges, reducing both shortage risk and excess inventory cost. This practical supply chain application is consistent with the use case highlighted by Çetin and Taşdemir (2024) for the US EV market.

6.3 Why SARIMA is the Preferred Model

Despite Holt-Winters achieving marginally lower RMSE on this particular test window, SARIMA is the preferred model for three reasons. First, SARIMA's selection via nested rolling CV provides a more robust validation than a single-split evaluation, making it less sensitive to the specific characteristics of one 12-month hold-out period. Second, SARIMA's parameters (AR, I, MA terms with seasonal analogs) have direct economic interpretability: the negative non-seasonal AR coefficient reflects short-term mean reversion in monthly sales, and the negative seasonal AR coefficient reflects the year-over-year market stabilization since 2022. Holt-Winters' smoothing parameters lack this structural interpretation (Ge et al., 2024). Third, SARIMA's explicit probabilistic framework naturally provides confidence intervals that quantify forecast uncertainty in a theoretically principled way, supporting the scenario-based planning described above (Çetin and Taşdemir, 2024). These three arguments echo the reasoning of Chen et al. (2025) and Liang (2025) for preferring SARIMA over purely exponential smoothing alternatives in the China NEV context.

6.4 Limitations

This paper's results should be interpreted with three important caveats.

Univariate model limitation: SARIMA relies entirely on lagged values of the target series. It cannot anticipate policy discontinuities (subsidy changes, purchase restrictions), macroeconomic shocks (income growth, fuel price changes, exchange rate movements), or competitive dynamics (new model launches, price wars). The January 2023 post-subsidy collapse and the early 2024 underperformance both demonstrate this fundamental limitation, which is acknowledged across the entire body of related EV forecasting literature (Chen et al., 2025; Ge et al., 2024; Yang, 2024; Liang, 2025). Yang (2024)'s regression analysis, which identifies charging infrastructure and battery output as $\rho = 0.99$ correlated with NEV demand, suggests that a SARIMAX extension incorporating these supply-side indicators could substantially reduce this gap.

Short sample and structural breaks: With only 76 monthly observations spanning two qualitatively different market regimes (pre- and post-subsidy phase-out), the model must simultaneously learn both the high-growth pattern of 2018–2022 and the more stable pattern of 2023–2024. This dual-regime estimation may reduce the model's ability to capture either regime cleanly, and SARIMA's assumption of stable seasonality over time may not hold across the full sample period. This limitation is less severe in the studies by Chen et al. (2025) and Liang (2025), which use shorter windows focused on more homogeneous market regimes, at the cost of reduced information for identifying structural parameters.

Aggregation: The analysis focuses on total EV sales without disaggregation by vehicle segment (BEV vs. PHEV), price tier, or brand. BEVs and PHEVs have distinct seasonal patterns and different sensitivities to policy changes; aggregate forecasting blends these dynamics and may mask important sub-market trends. Future disaggregated analysis would benefit from the factor-level framework of Ge et al. (2024), who decompose NEV demand across technology, policy, and market dimensions, combined with the segment-level data quality standards advocated by Yang (2024).

7 Conclusion

This paper applies a fully documented SARIMA framework to forecast monthly EV sales in China from a 76-month historical series spanning January 2018 to April 2024. The primary methodological contribution is a two-stage model selection procedure that combines BIC-guided auto-ARIMA with nested rolling cross-validation, producing a final specification of SARIMA$(1,1,0) \times (1,1,0)_{12}$ that is more parsimonious and more robust than the BIC-optimal in-sample model.

The key empirical findings are: (1) the China EV sales series is strongly non-stationary in levels but achieves confirmed stationarity after one non-seasonal and one seasonal difference, as validated by both ADF and KPSS tests, replicating the stationarity structure reported by Liang (2025) and Lu (2024) for their respective EV markets; (2) the selected SARIMA model achieves test-set MAPE of 17.42% and passes the Ljung-Box residual independence test at lag 12 ($p = 0.336$); (3) an additive specification on the original series significantly outperforms a multiplicative log-space alternative, due to the structural market shift in 2023 documented by Chen et al. (2025); (4) the 12-month forward forecast predicts a December 2024 peak of approximately 810,000 units and a February 2025 trough of approximately 446,000 units, consistent with the Q4 seasonal peak pattern established across all China-focused studies reviewed (Chen et al., 2025; Liang, 2025; Ge et al., 2024; Yang, 2024).

These forecasts have practical relevance for manufacturers, distributors, policymakers, and infrastructure planners. The December peak projection supports proactive Q3–Q4 inventory and production builds, while the February trough informs planned production slowdowns and seasonal cash flow management. The 95% confidence intervals facilitate low/high scenario planning across stakeholder groups. The infrastructure planning implications are particularly salient in the context of Chen et al. (2025)'s argument that SARIMA forecasts provide the scientific basis for staging charging network expansion, and Ge et al. (2024)'s finding that NEV sales growth Granger-causes policy effectiveness rather than the reverse.

Future work should extend this framework in three directions. First, incorporating exogenous policy variables – such as subsidy amounts, purchase restriction policies, and the charging infrastructure density metric highlighted by Yang (2024) – via SARIMAX would reduce the model's vulnerability to policy shocks. Second, disaggregating the forecast by BEV versus PHEV segment, or by major OEM, would provide more actionable granularity for industry decision-makers, building on the factor-level decomposition of Ge et al. (2024). Third, the model should be updated on a rolling basis as new monthly observations become available, with automated anomaly detection to flag structural breaks early – a natural extension toward the type of adaptive forecasting demonstrated by Lu (2024) in the UK context.

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A Appendix

A.1 Implementation Details

The entire pipeline was implemented in Python 3.11 using the following libraries: `pandas` for data loading and monthly aggregation; `numpy` for numerical operations; `matplotlib` and `seaborn` for visualization; `statsmodels` for SARIMAX estimation, ADF, KPSS, and Ljung-Box testing; `pmdarima` for auto-ARIMA and rolling cross-validation; `prophet` for the Facebook Prophet baseline; and `tensorflow/keras` for the LSTM baseline. The code is organized into two notebooks: `01_Data_Prep_and_EDA.ipynb` (preprocessing, decomposition, outlier analysis) and `02_Time_Series_Modeling.ipynb` (stationarity tests, model selection, diagnostics, forecast comparison). All random seeds are fixed at 42 for reproducibility.

A.2 SARIMA Algorithm Outline

Input: Monthly EV sales series $\{y_t\}_{t=1}^{T}$, seasonal period $m = 12$.
Step 1 — Data Preprocessing: Remove duplicates; convert `units_sold` to integer; filter EV records; aggregate to monthly start frequency; fill missing months with zero.
Step 2 — Exploratory Analysis: Plot raw series, compute summary statistics, perform additive decomposition to inspect trend, seasonal, and residual components. Apply 3-sigma outlier check on residuals.
Step 3 — Stationarity: Run ADF and KPSS on raw series and differenced variants. Select $(d, D) = (1, 1)$ based on dual test confirmation.
Step 4 — Initial Model Search (Stage 1): Run auto-ARIMA with $p, q \leq 2$, $P, Q \leq 1$, $d = 1$, $D = 1$, $s = 12$, BIC criterion, stepwise search. Record top-3 candidates by BIC.
Step 5 — Nested Rolling CV (Stage 2): For each fold in a rolling expanding window, re-fit auto-ARIMA and record one-step-ahead RMSE. Aggregate mean RMSE across folds per order configuration. Select the order with lowest mean RMSE.
Step 6 — Residual Diagnostics: Inspect residual ACF, histogram, Q-Q plot, and Ljung-Box $Q(12)$ statistic.
Step 7 — Baseline Comparison: Fit ARIMA, Holt-Winters, Prophet, and LSTM on the same training split. Evaluate all models on the 12-month test set.
Step 8 — Final Forecast: Refit selected SARIMA on all T observations. Generate $h = 12$ step-ahead forecasts with 95% confidence intervals.
Output: SARIMA parameter estimates, test-set RMSE/MAPE, residual test p-values, and the 12-month forecast table.